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Data Balancing in Breast Cancer Recognition Using the Wisconsin Breast Cancer Dataset

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***Abstract*— Breast cancer is ranked as the most diagnosed cancer among women around the world, and it is one of the main causes of death in women. Accurate and early detection of breast cancer is essential for improving patient outcomes, as it allows for timely treatment and better chances of survival. One major challenge for reliable detection is that the datasets used in medical diagnostics are often imbalanced, with far fewer malignant cases compared to benign ones, which may skew machine learning (ML) methods to yield inaccurate detections. This study explored various ways to use suitable ML classifiers such as Logistic Regression (LR), Random Forest (RF), Multilayer Perceptron Neural Network (MLP NN), Support Vector Machine (SVM), and Decision Tree (DT). Monte Carlo simulations were used to evaluate the recognition performance despite the challenge of data imbalance. The Wisconsin Breast Cancer Dataset (WBCD) is a benchmark database for ML research, with 212 malignant (cancerous) and 357 benign (non-cancerous) cases. From WBCD, an imbalanced dataset was randomly created to simulate real-world diagnostic conditions, and the results are compared with the "truth" when more minority cases were available. To address class imbalance, data balancing techniques such as SMOTE (Synthetic Minority Oversampling TEchnique) and weighted bootstrapping (with 30% and 50% weighting) were applied. The performance of the classifiers was then evaluated on both the imbalanced dataset and a separate cross-validation dataset. Additionally, each classifier's performance was compared against its own baseline on balanced data to evaluate the impact of the balancing techniques. Among all classifiers, Logistic Regression consistently achieved the best performance, maintaining over 99% accuracy and up to 99.34% recall on the imbalanced dataset, and achieving up to 95.77% accuracy and 93.12% recall on the cross-validation dataset. This makes LR the most suitable classifier for preliminary breast cancer recognition.**

Keywords*— *Breast Cancer Detection, Synthetic Minority Over-Sampling Technique (SMOTE), Weighted Bootstrapping, Monte Carlo Simulations, Machine Learning, Wisconsin Breast

I. INTRODUCTION

Breast cancer is the most commonly diagnosed cancer in women, and it is the second leading cause of cancer-related deaths (only after lung cancer). Studies show that approximately 1 in 10 women may be affected by breast cancer earlier or later in their lives. Breast cancer accounts for around 25% of all cancer cases and about 12% of new cases in women [1]. One important step in diagnosing breast cancer is identifying whether a tumor is benign or malignant, which could be a challenging task even for experienced doctors. To detect breast cancer, patients often go through several tests such as ultrasound, mammography, and biopsy, depending on their symptoms. Biopsy is the most reliable by extracting a small sample of tissue or cells, using Fine Needle Aspiration (FNA) technique, for detailed examination under a microscope [2]. These microscopic images can provide numerical variables such as radius, texture, perimeter, and area, which are then analyzed, often along with imaging data, to predict whether a tumor is likely to be malignant. Early diagnosis of breast cancer can greatly raise a patient's chances of survival and there is a strong need for a reliable and accurate automatic diagnostic system. However, cancer occurs less frequently than benign tumors, which creates a major challenge to develop reliable predictive methods using machine learning (ML) with such imbalanced training datasets. The Wisconsin Breast Cancer Dataset (WBCD) collected by the FNA biopsy technique has been a benchmark among researchers and it is not as imbalanced as in the real life, so an imbalanced dataset could be sampled from it to reflect the real-world diagnostic conditions while all the data could be used to establish a "truth" baseline to compare with the recognition results using data balancing methods, including SMOTE and weighted bootstrapping (with 30% and 50% weight on the minority class generation), on the imbalanced data. Therefore, this study partitioned WBCD to evaluate multiple ML algorithms using various data balancing methods on both the imbalanced dataset and a separate cross-validation dataset.

The objectives of this research include: (1) to simulate and handle data imbalance to detect breast cancer; (2) to identify the most effective classifier (out of LR, RF, MLP, SVM, and DT) for reliable breast cancer detection; and (3) to compare the classifiers' relative improvements against balanced baseline results to evaluate the effectiveness of balancing techniques. In the rest of this paper, Section II reviews work in breast cancer detection; Section III describes the methodology with dataset preparation and data balancing techniques. Section IV presents the evaluation process. Section V presents the experimental results, comparing the performance of various classifiers on both imbalanced and cross-validation datasets. Section VI discusses the key findings and implications. Then Section VII concludes the study and outlines directions for future research.

II. LITERATURE REVIEW

Cancer diagnoses are often imbalanced with significantly fewer malignant cases than benign ones, leading to misclassification by ML. Previous studies have proposed

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various methods [3] that use different feature sets and machine learning methods to diagnose breast cancer using WBCD. For example, R. Fitrah Umami and R. Sarno [4] implemented Generalized Linear Model (GLM), LR, and Gradient Boosted DT to detect breast cancer, with GLM achieving a high accuracy of 99.4%. S. Ara, A. Das, and A. Dey [5] applied SVM, LR, k-Nearest Neighbors (KNN), DT, Naive Bayes (NB), and RF, to classify breast tumors as benign or malignant using the WBCD, with RF and SVM achieving the highest performance, both reaching an accuracy of 96.5%. Similarly, A. Rovshenov and S. Peker [6] employed Artificial Neural Networks (ANN), SVM, and RF algorithms to classify cancers using the WBCD, with ANN model achieving the highest accuracy of 99%. M. I. H. Showrov, M. T. Islam, M. D. Hossain, and M. S. Ahmed [7] conducted a study on WBCD with SVM, ANN, NB to find that the Linear SVM model achieved the highest accuracy of 96.72%. C. Dubey, N. Shukla, D. Kumar, A. K. Singh, and V. K. Dwivedi [8] applied eight machine learning algorithms (SVM, LR, XGBoost, CatBoost, RF, ANN, DT, and KNN) to the WBCD with SVM and ANN delivering the highest accuracy of 98.08%. However, all these studies only focused on accuracy but not on precision, recall, or F1 score, which are essential for model performance evaluation when false detection has detrimental consequences.

There are other studies on various optimization techniques in ML. For instance, N. Darapureddy, N. Karatapu and T. K. Battula [9] developed a deep neural network and implemented RMSprop and Stochastic Gradient Descent (SGD) to support computer-aided diagnosis and improved the effectiveness of radiologists over traditional screening. V. Gnanaprakash, K. Shoukath Ali, D. Preethi, N. Saranya, and K. Nivetitha R [10] developed a Convolutional Neural Network (CNN) model to classify tumors using WBCD, achieving 99% accuracy. They compared the performance of different optimizers and found that the Adam optimizer outperformed others to yield favorable accuracy, recall, precision, and F1 score for both CNN methods with and without a dropout layer.

Feature extraction would be impactful, too. Y. Ono and Y. Mitani [11] developed a computer-aided diagnosis system and highlighted the effectiveness of the RF classifier, particularly when combined with Recursive Feature Elimination (RFE), showing better performance than other ensemble methods like XGBoost and LightGBM. S. C. S. R., H. Rajaguru, and S. Chidambaram [12] applied the Ebola Optimization Algorithm (EOA) for feature selection and used a mixture Kernel Support Vector Machine (mK-SVM) for classification, and linear SVM and KNN, achieving a maximum classification accuracy of 97.19%. However, these studies did not address the issue of class imbalance that would happen in real life.

Using mostly other databases, several researchers have applied data balancing techniques in this field. For example, F. Y. A'la, N. Firdaus, Hartatik and M. A. Safi'Ie, [13] addressed class imbalance in the Coimbra breast cancer dataset using SMOTE and built a RF model evaluated with 10-fold cross-validation. Their approach improved model accuracy from 76.72% to 80.47%, precision from 76.60% to 80.00%, and recall from 69.23% to 81.25%, enhancing the model's ability to predict outcomes based on clinical and demographic features. M. M. Rahman, S. Akther, K. H. Kobir, T. Islam, A. Mahmud, and T. I. Efa [14] proposed an ensemble-based approach for breast cancer prediction using SMOTE and recursive feature elimination. They compared multiple algorithms, including LR, RF, SVM, and Gradient Boosting, with hyperparameter tuning via grid search. Their voting classifier achieved outstanding accuracy of 99.55% and 99.46% on WBCD and WBC datasets. They applied SMOTE to address the class imbalance, although their datasets were not highly imbalanced. N. I. B. Ali, S. A. M. Jamil, N. Ibrahim, and N. S. M. Zuki [15] conducted a survival analysis of breast cancer patients in Johor, Malaysia, using a small dataset of 32 cases split by age groups. They applied bootstrapping techniques to overcome sample size limitations and used the Kaplan-Meier method for survival estimation.

While previous studies have contributed significantly to breast cancer classification, many focused on limited feature sets, specific algorithms, or limited data-balancing techniques. This study used the WBCD to construct an imbalanced dataset mimicking real-world diagnostic conditions. We applied LR, RF, MLP, SVM, and DT, alongside SMOTE and weighted bootstrapping (with 30% and 50% weight on the minority class generation). The methods were then evaluated on both the imbalanced dataset and a separate cross-validation dataset to assess their generalization capability. This study not only evaluated classifier performance but also compared it against the balanced baseline results to highlight the impact of different balancing techniques.

III. METHODOLOGY

A. Data pre-processing

This study used the publicly available Wisconsin Breast Cancer Dataset (WBCD), developed by Dr. William H. Wolberg at the University of Wisconsin Hospital in Madison, USA [16]. The dataset includes 32 columns, where the first column is the ID number and the second column is the diagnosis outcome. The remaining 30 columns are three groups of statistics: mean, standard error, and worst (largest) mean values, of ten morphological variables listed in TABLE I. . These variables describe the size and shape characteristics of the cell nuclei, observed from microscopic images of breast cell samples obtained through FNA biopsy. All variables values are recorded with four significant digits, with no missing values.

TABLE I. VARIABLES IN WBCD [16]

Variables	Description
Radius	Mean distance from the center to the boundary points.
Texture	Standard deviation of gray-scale values.
Perimeter	Total distance around the cell boundary (snake points).
Area	Number of pixels inside the boundary, including half the edge pixels.
Smoothness	Local variance in radius lengths measures irregularity.
Compactness	Calculated as (Perimeter^2/Area).
Concavity	Measures the severity of inward curves on the cell contour.
Concave points	Number of distinct concave regions on the contour.
Symmetry	Difference in length of perpendicular lines to the major axis on both sides of the boundary.
Fractal Dimension	Complexity of the boundary shape; higher values indicate greater irregularity, linked to malignancy risk.

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The ranges of these variable values are shown in TABLE II. As the variables have different numerical ranges, all variables were standardized to have a unit normal distribution.

TABLE II. WBCD'S DATA RANGES

	Average Values across the Database					
For individuals:	Mean		Standard Deviation		Worst Case	
Variables	**Benign**	**Cancerous**	**Benign**	**Cancerous**	**Benign**	**Cancerous**
Radius	12.147	17.463	0.284	0.609	13.380	21.135
Texture	17.915	21.605	1.220	1.211	23.515	29.318
Perimeter	78.075	115.365	2.000	4.324	87.006	141.370
Area	462.790	978.376	21.135	72.672	558.899	1422.286
Smoothness	0.092	0.103	0.007	0.007	0.125	0.145
Compactness	0.080	0.145	0.021	0.032	0.183	0.375
Concavity	0.046	0.161	0.026	0.042	0.166	0.451
Concave Points	0.026	0.088	0.010	0.015	0.074	0.182
Symmetry	0.174	0.193	0.021	0.020	0.270	0.323
Fractal Dimension	0.063	0.063	0.004	0.004	0.079	0.092

a) Imbalanced dataset

The WBCD was a relatively balanced dataset with 569 samples: 212 malignant and 357 benign cases. However, to reflect real-world scenarios where cancer cases are much less frequent compared to non-cancer cases, we randomly sampled an imbalanced dataset from the WBCD consisting of 287 samples: 43 malignant and 244 benign cases. This imbalanced dataset was used for training and testing the proposed methods.

b) Cross-Validation dataset

The rest of the samples that were not used in training/testing were used as the cross-validation dataset, with 169 malignant and 113 benign cases. This dataset is unseen by the algorithms and unchanged, to test the generalizability of the algorithms.

B. Handle Imbalanced data

a) *SMOTE (Synthetic Minority Over-Sampling* TEchnique)

SMOTE [17] is a technique to generate synthetic samples for the minority class based on the distribution of the classes, helping machine learning methods learn more effectively from both classes. The imbalanced dataset comprised of 287 rows, with 14.98% belonging to the malignant class and 85.01% to the benign class. After applying SMOTE using the imblearn library in Python 3.11 with random_state=42 and the default parameter K_neighbors=5, synthetic samples were created for the malignant class (minority), producing a balanced dataset of 488 instances, with approximately equal representation of malignant and benign cases, as shown in the Fig. 1.

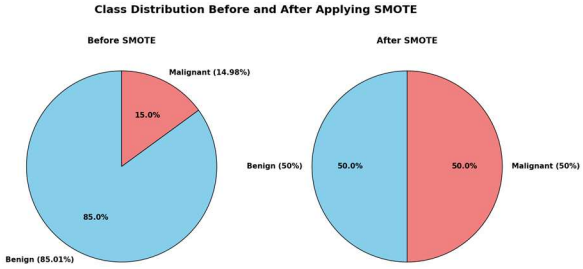

Fig. 1. Distribution of classes before and after SMOTE oversampling.

b) Weighted bootstrapping

Another approach to handle imbalanced class distributions is weighted bootstrapping [18]. We applied a 30% weight, initially, to the malignant class, to yield approximately 30% malignant samples and 70% benign samples. After applying weighted 30% bootstrapping using Python 3.11 and pandas, with random_state=42 and sampling with replacement, the new dataset consisted of 574 samples, with the proportion of malignant cases increasing to 29.96% and benign cases decreasing to 70.03%, as illustrated in Fig. 2.

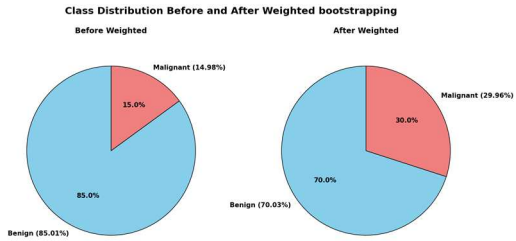

Fig. 2. Classs balance before and after using 30% weighted bootstrapping on the minority group.

Next, a weighted bootstrapping with a 50% weight was used to yield roughly equal numbers of benign and cancerous cases.

IV. MODEL EVALUATION PROCESS

A. Data splitting

As shown in Fig.3., the data was partitioned into 80% training and 20% testing sets for the imbalanced dataset and those generated datasets using SMOTE and two weighted bootstrapping. For instance, out of 287 imbalanced samples, 229 were randomly chosen for training and the other 58 were for testing. This random partitioning was repeated in each Monte Carlo run, ensured proper evaluation of the methods' ability to accurately detect breast cancer. Lastly, the trained methods were assessed using cross-validation dataset on their generalization ability.

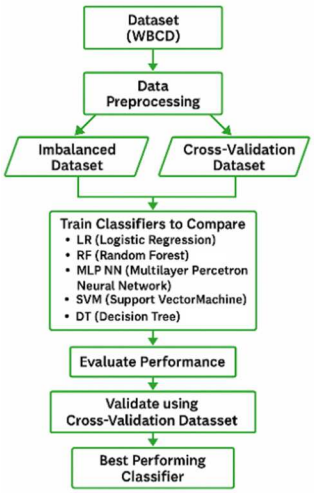

Fig. 3. Step-by-step schematic of the model evaluation procedure.

B. Algorithm selection and model training

This study evaluated Logistic Regression (LR) [19], Random Forest (RF) [20], Multilayer Perceptron Neural Network (MLP NN) [21], Support Vector Machines (SVM) [22], and Decision Tree (DT) [23] to determine the most effective model for breast cancer detection. Parameters were optimized to minimize recognition errors on the training data.

C. Model evaluation

A confusion matrix records the classification results compared with the truth in four quadrants: TP represents the number of true positive samples, TN represents true negatives, FP represents false positives, and FN represents false negatives. Then several metrics are calculated. Accuracy measures the ratio

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of correct classification by (1):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Precision is the proportion of TP among all positive predictions by (2):

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

Recall is the proportion of TP among all true positive decisions by (3):

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

F1 score combines precision and recall, providing a balanced measure that combines both in (4):

$$\text{F1 Score} = 2 \cdot \frac{precision \cdot recall}{precision + recall} \quad (4)$$

For breast cancer detection, minimizing false negatives is critical, as failing to identify a malignant case has serious consequences. Therefore, accuracy and recall are emphasized to ensure that cancer cases are not overlooked.

V. Numerical Analysis

This study analyzes the performance of multiple classifiers in detecting malignant and benign breast cancer cases alongside data balancing techniques to the imbalanced dataset, including SMOTE and weighted bootstrapping. Cross-validation is used to assess their generalization capability.

In Fig. 4, the top row shows results from testing the imbalanced dataset after training, while the bottom row shows results from cross-validation. Accuracy and recall are presented in the left and right columns, respectively, with the X-axis starting from the baseline performance on the imbalanced dataset, followed by results from the data-balancing methods from left to right for comparison. Each line represents a different classifier, distinguished by unique line styles and colors. On the imbalanced dataset shown in the top, the LR classifier (the line in green with square markers) achieved the optimal performance, with the highest accuracy of 99.09% and recall of 94.67%, and it outperformed all other classifiers. With data balancing techniques, LR continued to deliver consistently good performance in terms of accuracy and recall. With the data balancing techniques on imbalanced dataset, the MLP classifier (the line in orange with cross markers) outperformed LR (represented by the line in green with square markers), but the difference was minimal. RF (line in blue with diamond markers) also performed effectively, especially in recall (up to 99.79%) and accuracy (up to 99.89%). SVM showed consistent improvement from 98.52% to 99.54% accuracy and from 91.44% to 99.34% recall. However, DT had the lowest baseline performance, with 95.64% accuracy and 87.78% recall on the imbalanced dataset, though it improved significantly with data balancing.

On the cross-validation dataset in the bottom, LR achieved the highest recall of 93.12% and maintained high accuracy up to 95.77%, with or without applying the data balancing techniques. In contrast, DT consistently showed the lowest performance. Although RF and SVM showed moderate accuracy and recall, their performance was not as consistent as LR or MLP.

Data balancing techniques were experimented on the randomly sampled imbalanced dataset as explained earlier, and since the WBCD does include more samples of the minority class, a baseline analysis on a balanced dataset is done by choosing the same number of malignant and benign samples. The results on balanced data are shown in TABLE III. .

TABLE III. Baseline Performance from Balanced Data

Algorithms	Accuracy	Recall
LR	0.9576	0.9534
RF	0.9469	0.9510
MLP	0.9591	0.9448
SVM	0.9560	0.9517
DT	0.9021	0.9121

TABLE III. shows that the baseline performance of different algorithms differs, so each algorithm in Fig. 4 is compared with its own baseline performance from TABLE III. , to be presented in Fig. 5, where the thick dashed line is each algorithm's baseline, treated as the common zero-line. On the testing dataset during the training phase (the upper row in Fig. 5), the accuracy is way over the baseline, and the recall after data-balancing is above the baseline. On the cross-validation dataset (the lower row in Fig. 5), the accuracy and recall on the imbalanced dataset is lower than the baseline. Still, LR using either data balancing technique is approaching the baseline, while the recall rate drops about 5%. This follow-up study illustrates that data balancing techniques help the performance on imbalanced data to approach the performance on balanced data, and LR algorithm is the best performing in this case.

TABLE IV. and TABLE V. detailed all the numeric metrics (Accuracy, Recall, Precision, F1-scoe, and AUC) of five ML models across different data balancing methods on both the testing and the cross-validation datasets. The AUC (Area under the ROC Curve) is from roc_auc_score() of sklearn.metrics. Overall, LR and SVM consistently achieved high scores. Balancing the data helps the models make better predictions, and LR is the most reliable for this task.

VI. Conclusions and Future Work

This study demonstrated the importance of appropriate classifier and data balancing technique to improve breast cancer detection on imbalanced datasets. Data balancing techniques enhanced recognition by improving malignant cases detection and reducing bias toward the benign cases. LR provided the most consistent and reliable results, especially on the cross-validation dataset. Early diagnosis supports timely treatment, leading to better patient outcomes and more efficient use of medical resources. Future research will explore interpretability of the LR model and examine if other ML techniques may lead to further performance improvements and if dimensionality reduction methods may impact accuracy and recall. This study used a relatively small dataset. We plan to test the model on larger or image-based datasets and explore the use of deep learning methods to improve performance further.

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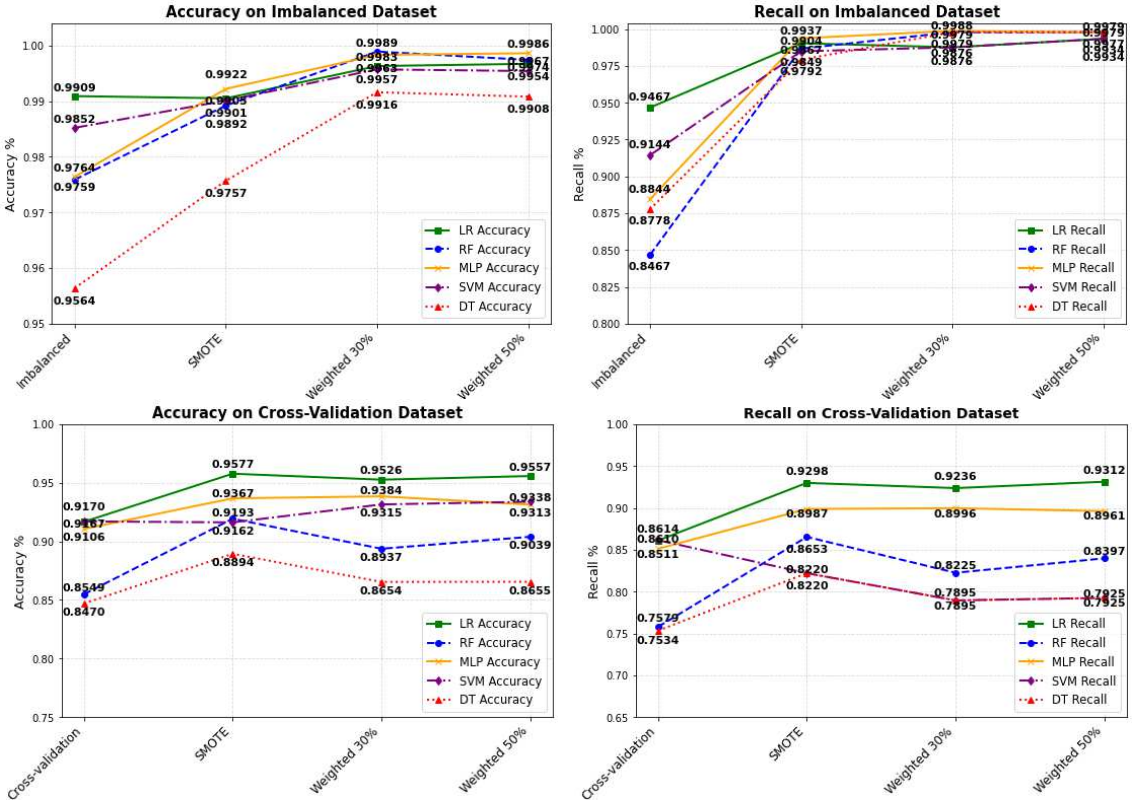

Fig. 4. Classifier performance in terms of accuracy and recall on imbalanced data (top) and cross-validation data (bottom)

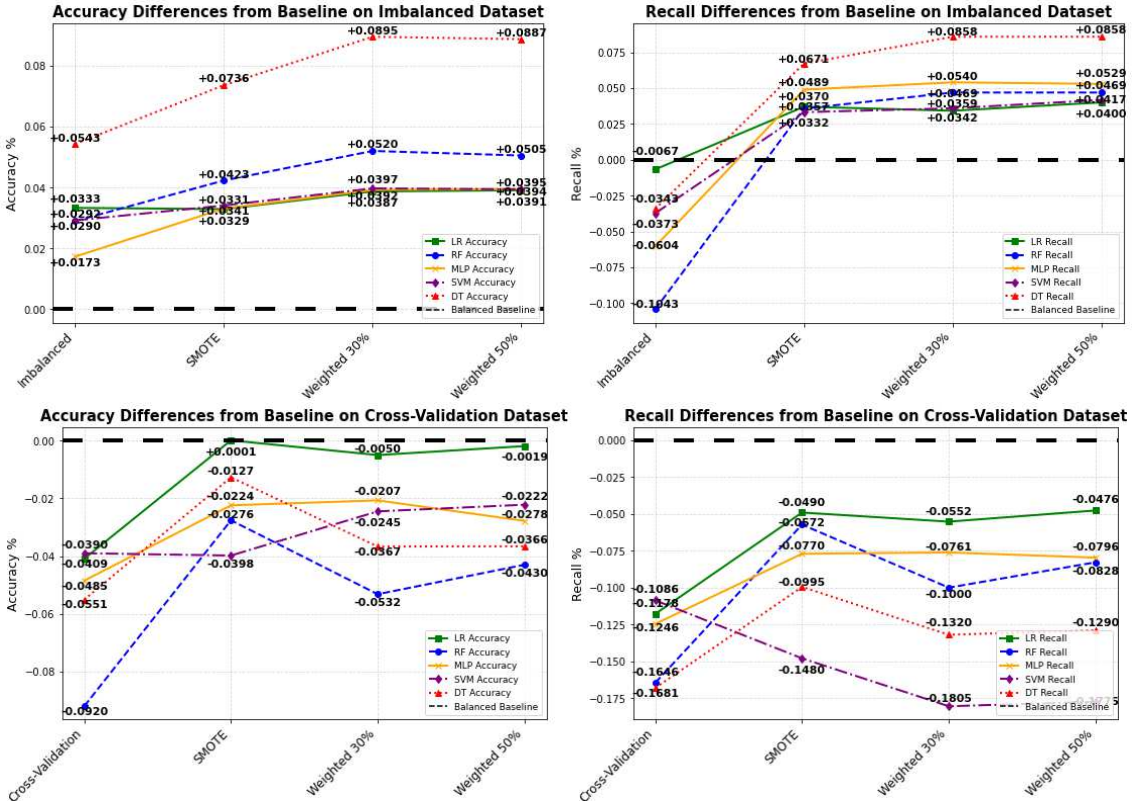

Fig. 5. Relative accuracy (left) and recall (right) improvements of classifiers across different data balancing techniques, compared to their own baseline performance on imbalanced data (top) and cross-validation data (bottom).

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TABLE IV. PERFORMANCE ON IMBALANCED TESTING DATASET WITH DIFFERENT TECHNIQUES

Average	Classifier	Imbalanced Data	SMOTE	Weighted 30% bootstrapping	Weighted 50% bootstrapping
Accuracy	LR	0.9909	0.9905	0.9963	0.9967
	RF	0.9759	0.9892	0.9989	0.9974
	MLP	0.9764	0.9922	0.9983	0.9986
	SVM	0.9852	0.9901	0.9957	0.9954
	DT	0.9564	0.9757	0.9916	0.9908
Recall	LR	0.9467	0.9904	0.9876	0.9934
	RF	0.8467	0.9867	0.9979	0.9979
	MLP	0.8844	0.9937	0.9988	0.9977
	SVM	0.9144	0.9849	0.9876	0.9934
	DT	0.8778	0.9792	0.9979	0.9979
Precision	LR	0.9949	0.9908	1.0000	1.0000
	RF	0.9971	0.9918	0.9983	0.997
	MLP	0.9648	0.991	0.9958	0.9995
	SVM	0.9909	0.9953	0.9978	0.9974
	DT	0.859	0.9731	0.975	0.9843
F1 score	LR	0.9685	0.9905	0.9937	0.9967
	RF	0.9115	0.9891	0.9981	0.9974
	MLP	0.9188	0.9923	0.9972	0.9986
	SVM	0.9483	0.99	0.9926	0.9954
	DT	0.8618	0.9758	0.9861	0.9909
AUC	LR	0.9979	0.9997	0.9996	0.9998
	RF	0.9874	0.9987	0.9992	0.9988
	MLP	0.975	0.9998	0.9999	0.9997
	SVM	0.9972	0.9992	0.9993	0.9996
	DT	0.9243	0.9757	0.9934	0.9908

TABLE V. PERFORMANCE ON CROSS-VALIDATION DATASET WITH DIFFERENT TECHNIQUES

Average	Classifier	Imbalanced Data	SMOTE	Weighted 30% bootstrapping	Weighted 50% bootstrapping
Accuracy	LR	0.9167	0.9577	0.9526	0.9557
	RF	0.8549	0.9193	0.8937	0.9039
	MLP	0.9106	0.9367	0.9384	0.9313
	SVM	0.9170	0.9162	0.9315	0.9338
	DT	0.8470	0.8894	0.8654	0.8655
Recall	LR	0.8610	0.9298	0.9236	0.9312
	RF	0.7579	0.8653	0.8225	0.8397
	MLP	0.8511	0.8987	0.8996	0.8961
	SVM	0.8614	0.8601	0.8857	0.8895
	DT	0.7534	0.8220	0.7895	0.7925
Precision	LR	1.0000	0.9995	0.9972	0.9946
	RF	1.0000	1.0000	1.0000	1.0000
	MLP	0.9998	0.9953	0.9974	0.9883
	SVM	1.0000	1.0000	1.0000	1.0000
	DT	0.9886	0.9922	0.9824	0.9791
F1 score	LR	0.9252	0.9634	0.9589	0.9618
	RF	0.8620	0.9277	0.9026	0.9128
	MLP	0.9193	0.9445	0.9459	0.9399
	SVM	0.9254	0.9248	0.9394	0.9415
	DT	0.8545	0.8986	0.8750	0.8752
AUC	LR	0.9939	0.9935	0.9920	0.9927
	RF	0.9907	0.9932	0.9898	0.9912
	MLP	0.9845	0.9927	0.9905	0.9911
	SVM	0.9947	0.9942	0.9946	0.9946
	DT	0.8701	0.9061	0.8842	0.8836

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